

Class #2 Truth & Logical Equivalence

Big Picture Goal:

Relate developing your understanding of logic to the way your understanding of math has developed in the past.

Class goals:

1. Construct truth tables
2. Determine logical equivalence of statements
3. Determine if a set of connectives is adequate.

Big Picture:

How has your understanding of math progressed over the years?
What was the first thing learned that you called math?

Objects:

numbers

Operations:

$+$, $-$, $\times$, $\div$

} Arithmetic

Classification:

Positive, negative, integer, rational
special numbers like 0 and 1

Rules:

$a=b$, $a+b=b+a$, $a(b+c)=ab+ac$
 $(a+b)+c=a+(b+c)$ } Algebra

Relations between numbers:

Functions

Then functions became the objects and you went through the process again.

Start this process now with logic.

Objects: *Sentences*

Operations: *Connectives* $\sim, \wedge, \vee, \rightarrow, \leftrightarrow$

New twist: These objects have two possible values: *True and False*

Truth Tables are used to exhibit the truth value of sentences for given truth values of the basic variables making up the sentence.

The following truth tables define the values of the most basic sentences.

p	$\sim p$	p	q	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$
T	<i>F</i>	T	T	<i>T</i>	<i>T</i>	<i>T</i>	<i>T</i>
F	<i>T</i>	T	F	<i>F</i>	<i>T</i>	<i>F</i>	<i>F</i>
		F	T	<i>F</i>	<i>T</i>	<i>T</i>	<i>F</i>
		F	F	<i>F</i>	<i>F</i>	<i>T</i>	<i>T</i>

Explain truth values for $p \rightarrow q$.

If I win the lottery then I will quit my job.

p $\rightarrow$ *q*

<i>p</i>	<i>q</i>	<i>Did I lie?</i>	
T I win	T I quit	NO	T
T I win	F I don't quit	Yes	F
F I don't win	T I quit	NO	T
F I don't win	F I don't quit	NO	T

Three classifications of sentences:

Contingency may be either T or F $p \wedge q$

Tautology Always true $p \vee (\neg p)$

Contradiction Always False $p \wedge (\neg p)$

Example: Classify the sentence $(p \rightarrow q) \vee (q \rightarrow p)$

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \vee (q \rightarrow p)$
T	T	T	T	T
T	F	F	T	T
F	T	T	F	T
F	F	T	T	T

Tautology

What does it mean to say two sentences are logically equivalent?

A: If f is differentiable then f is continuous
 $\text{diff} \rightarrow \text{cont} \quad p \rightarrow q$

B: If f is not cont then f is not diff.
 $\sim \text{cont} \rightarrow \sim \text{diff} \quad (\sim q) \rightarrow (\sim p)$

A is equivalent to B.

B is called the contrapositive of A.

Make a truth table containing both $p \rightarrow q$ and $(\sim q) \rightarrow (\sim p)$

p	q	$p \rightarrow q$	$\sim q$	$\sim p$	$(\sim q) \rightarrow (\sim p)$
T	T	T	F	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Two statements are logically equivalent if they have the exact same column of truth values in the truth table.

Notation: $p \rightarrow q \equiv (\sim q) \rightarrow (\sim p)$

Find a sentence that is logically equivalent to $p \rightarrow q$ but uses only the connectives $\sim, \wedge, \vee$

p	q	$p \rightarrow q$	$(\sim p) \vee q$	$\sim (p \wedge (\sim q))$
T	T	T		
T	F	F		
F	T	T		
F	F	T		

This says we really don't need the $\rightarrow$ connective since we could always replace $p \rightarrow q$ by $(\neg p) \vee q$

Big Question: What is the smallest set of connectives we really need to make every possible truth table? What is an Adequate Set

Definition: A set of connectives is adequate if every possible truth column can be constructed by a sentence using the connectives in the set.

Don't need $\rightarrow$ since we can replace $\rightarrow$ using $\neg$ and $\vee$

The handout on the next page should convince you that the set $\{\neg, \wedge, \vee\}$ is adequate to create any truth column.

Adequate Set of Connectives

Find a sentence involving p, q, and r that is equivalent to each target using only the connectives $\sim$, $\wedge$, $\vee$.

p	q	r	Target A	Target B	Target C	Target D	Yours
T	T	T	F	T	T	T	
T	T	F	F	T	T	F	
T	F	T	F	F	F	F	
T	F	F	T	F	T	T	
F	T	T	F	F	F	F	
F	T	F	F	F	F	F	
F	F	T	F	F	F	T	
F	F	F	F	F	F	F	

Sentence A: $(p \wedge \sim q) \wedge (\sim r)$

Sentence B: $(p \wedge q \wedge r) \vee (p \wedge q \wedge \sim r) \equiv p \wedge q$

Sentence C: $(p \wedge q \wedge r) \vee (p \wedge q \wedge \sim r) \vee (p \wedge (\sim q) \wedge \sim r)$

Sentence D: $(p \wedge q \wedge r) \vee (p \wedge (\sim q) \wedge \sim r) \vee (\sim p) \wedge (\sim q) \wedge r$

Yours:

Can you formulate a strategy to hit any target?

Read about disjunctive normal
form